dl.py - C:\Users\pings\Desktop\dl.py (3.14.2)

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```
#Exp No:9 Experiment to understand Matplotlib library use cases in Data Science through visualization
#Description: Use Iris data set to understand the Matplot lib library

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 200)
# build Iris_Dataset (same as the csv in the manual)
iris = load_iris()
data = pd.DataFrame(iris.data, columns=['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm'])
data.insert(0, 'Id', range(1, 151))
data['variety'] = ['Iris-' + iris.target_names[i] for i in iris.target]
# two rows differ in the csv version used in the manual
data.loc[34, ['SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']] = [3.1, 1.5, 0.1]
data.loc[37, ['SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']] = [3.1, 1.5, 0.1]
print(data)
data.info()
print(data.describe())
print(data.value_counts('variety'))
sns.countplot(x='variety', data=data)
plt.show()
FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)
print(FinalDataset.head())
sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)
plt.show()
sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)
plt.show()
sns.pairplot(data, hue='variety', height=3)
plt.show()
for col in ['PetalLengthCm', 'PetalWidthCm', 'SepalLengthCm', 'SepalWidthCm']:
    sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, col).add_legend()
    plt.show()
```

Ln: 8 Col: 38

IDLE Shell 3.14.2

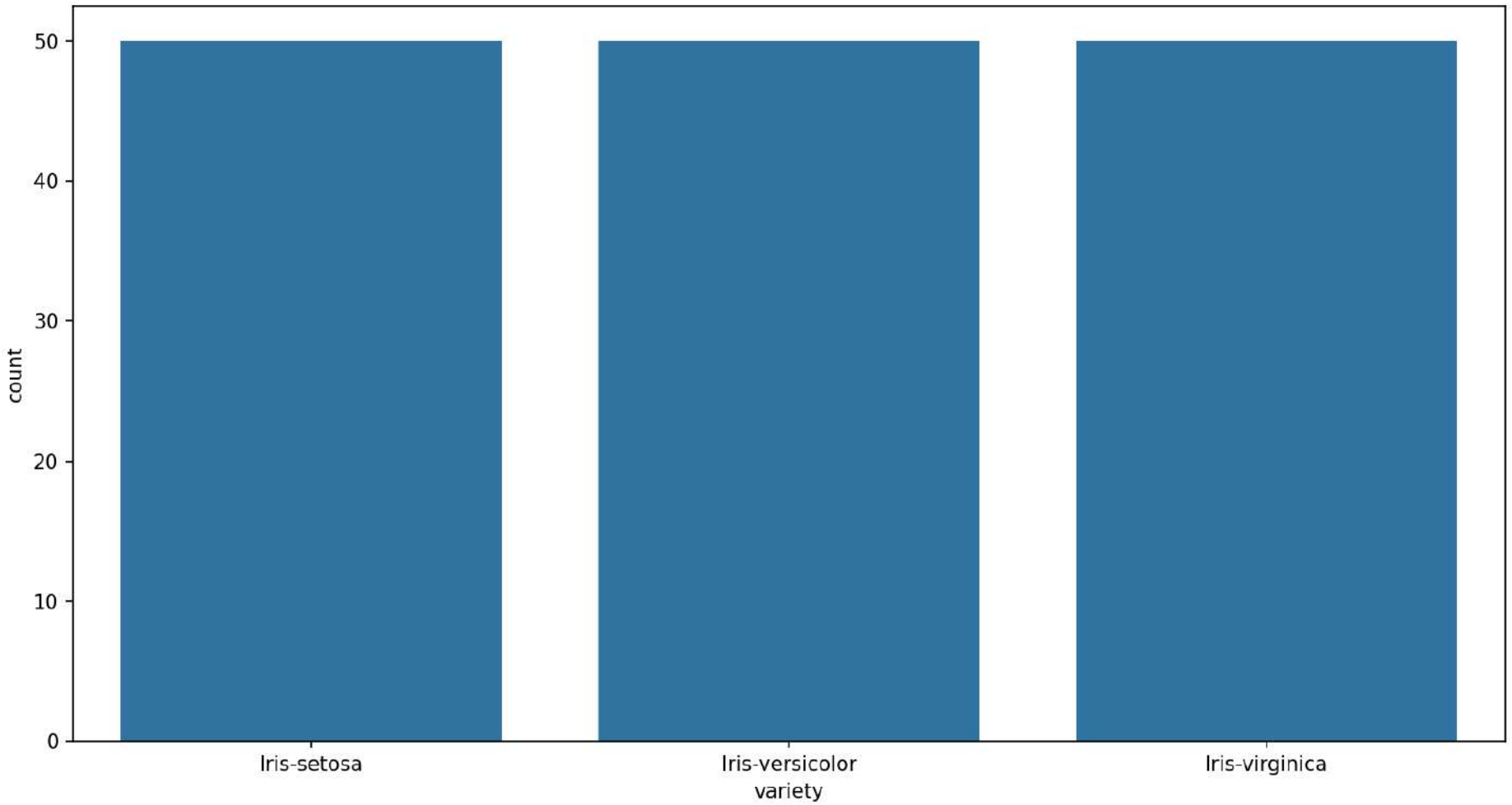
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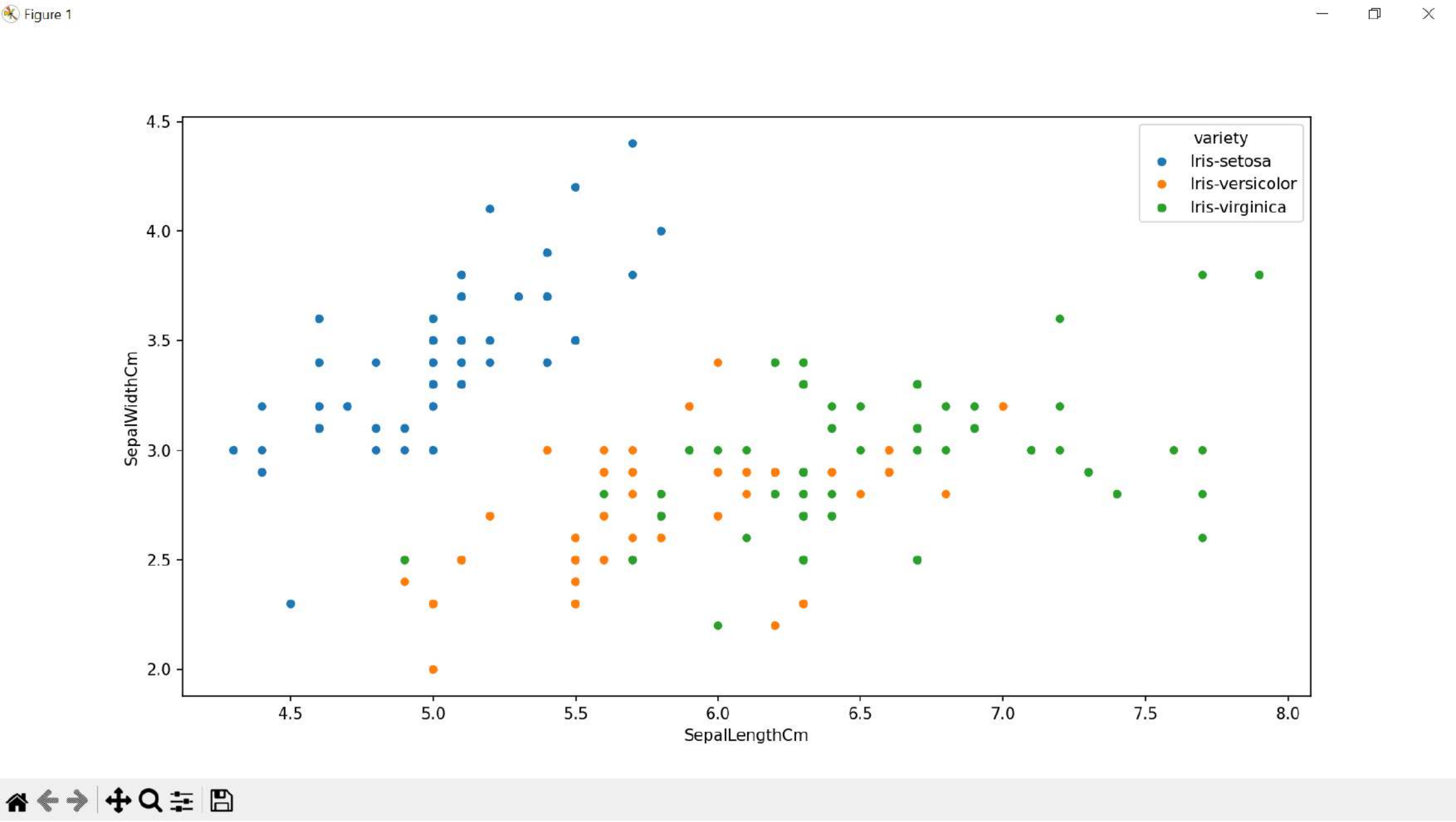
```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\pings\Desktop\dl.py ==========
      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm         variety
0      1            5.1           3.5            1.4           0.2     Iris-setosa
1      2            4.9           3.0            1.4           0.2     Iris-setosa
2      3            4.7           3.2            1.3           0.2     Iris-setosa
3      4            4.6           3.1            1.5           0.2     Iris-setosa
4      5            5.0           3.6            1.4           0.2     Iris-setosa
..   ...            ...           ...            ...           ...             ...
145  146            6.7           3.0            5.2           2.3  Iris-virginica
146  147            6.3           2.5            5.0           1.9  Iris-virginica
147  148            6.5           3.0            5.2           2.0  Iris-virginica
148  149            6.2           3.4            5.4           2.3  Iris-virginica
149  150            5.9           3.0            5.1           1.8  Iris-virginica

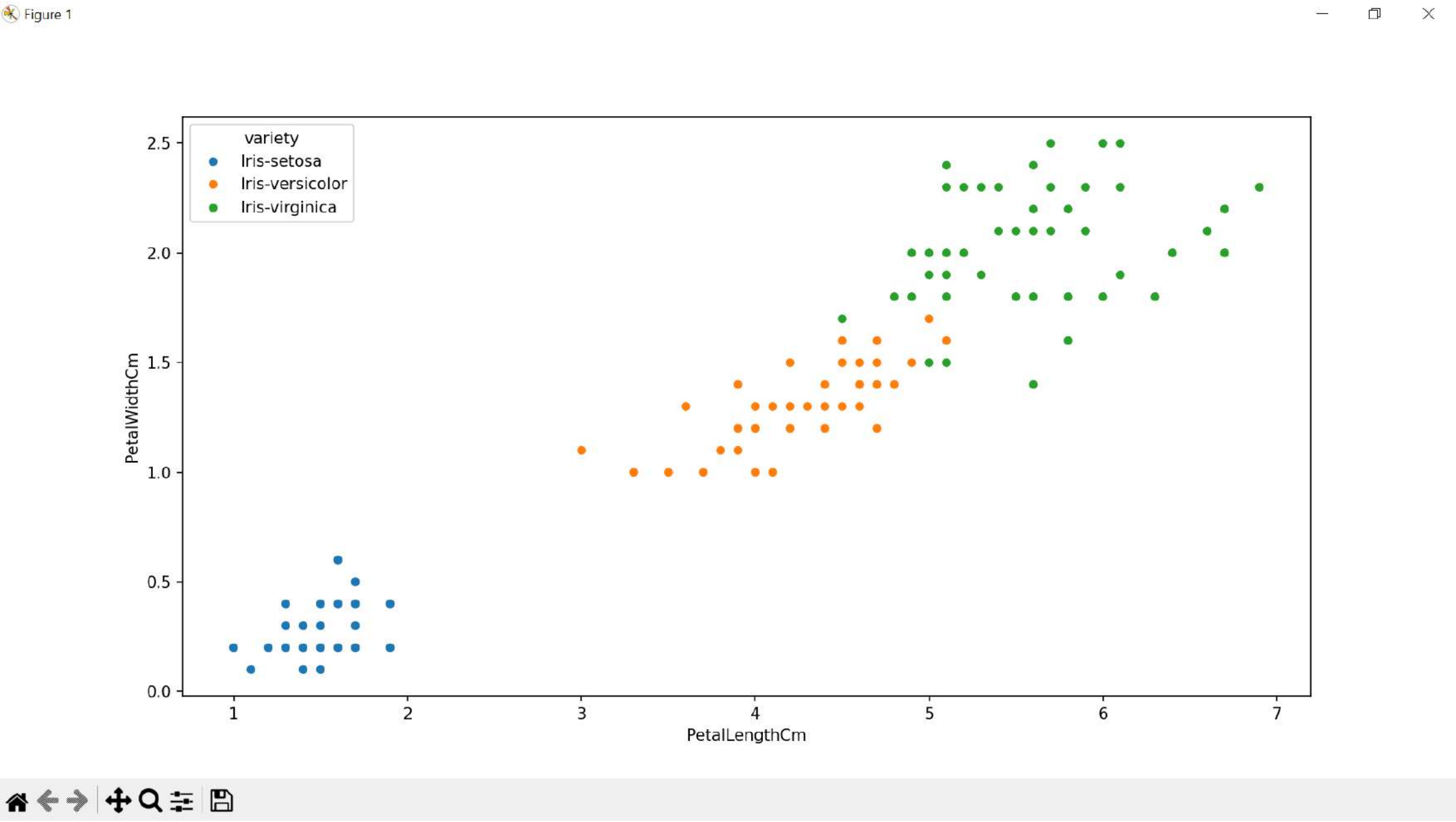
[150 rows x 6 columns]
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 9.1 KB
               Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
count  150.000000     150.000000    150.000000     150.000000    150.000000
mean    75.500000       5.843333      3.054000       3.758667      1.198667
std     43.445368       0.828066      0.433594       1.764420      0.763161
min      1.000000       4.300000      2.000000       1.000000      0.100000
25%     38.250000       5.100000      2.800000       1.600000      0.300000
50%     75.500000       5.800000      3.000000       4.350000      1.300000
75%    112.750000       6.400000      3.300000       5.100000      1.800000
max    150.000000       7.900000      4.400000       6.900000      2.500000
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

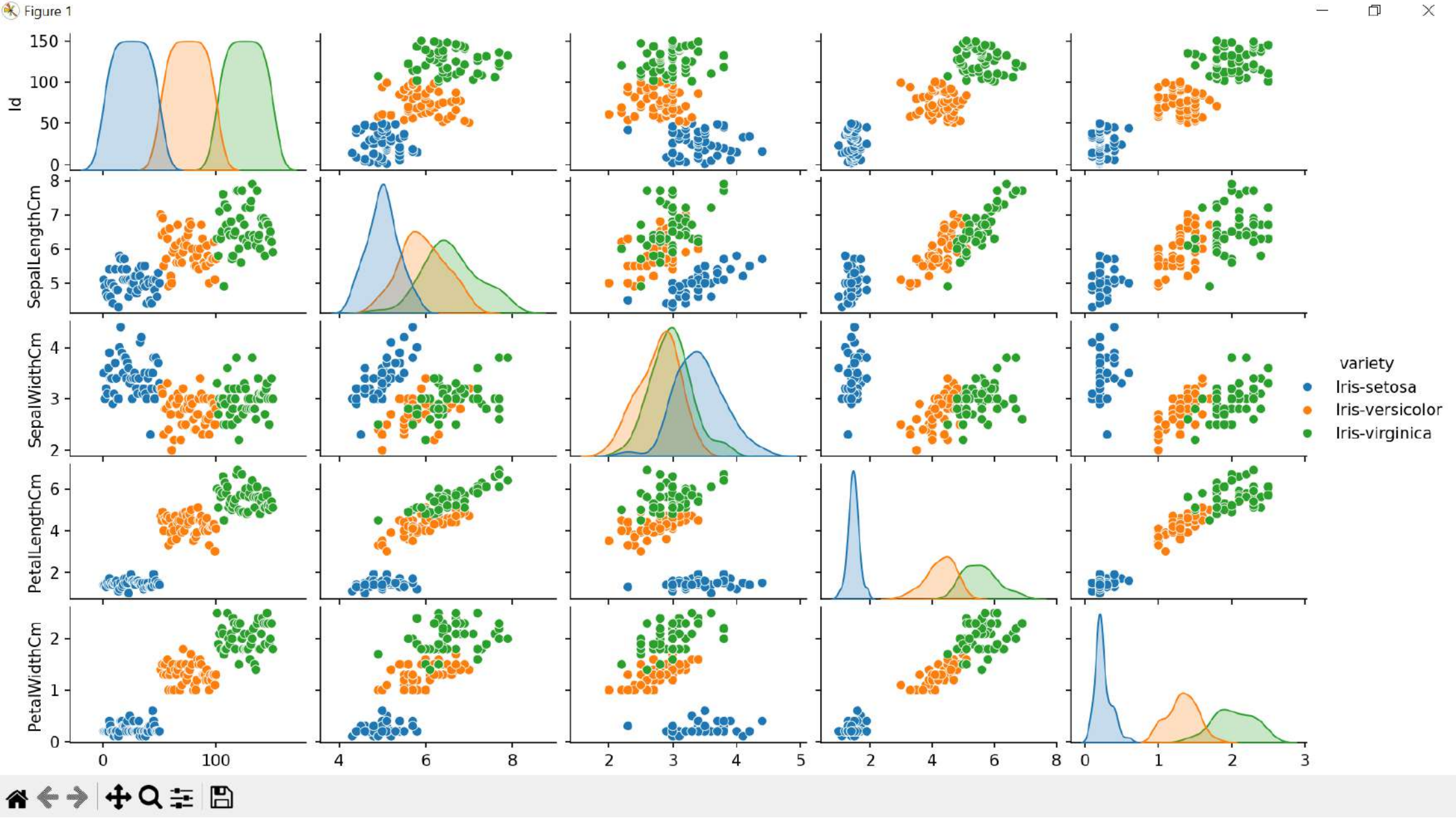
Ln: 5 Col: 0

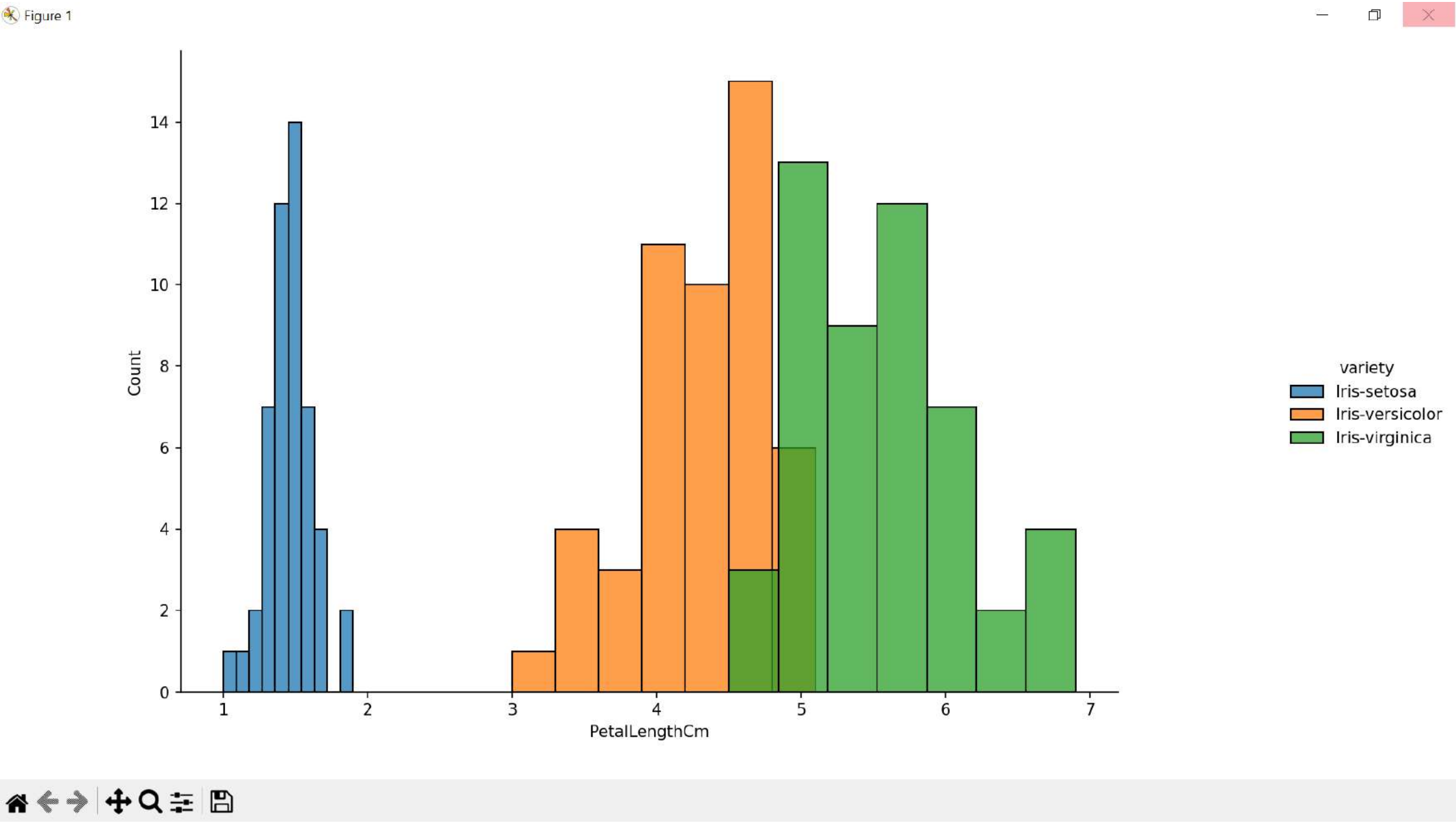
Figure 1

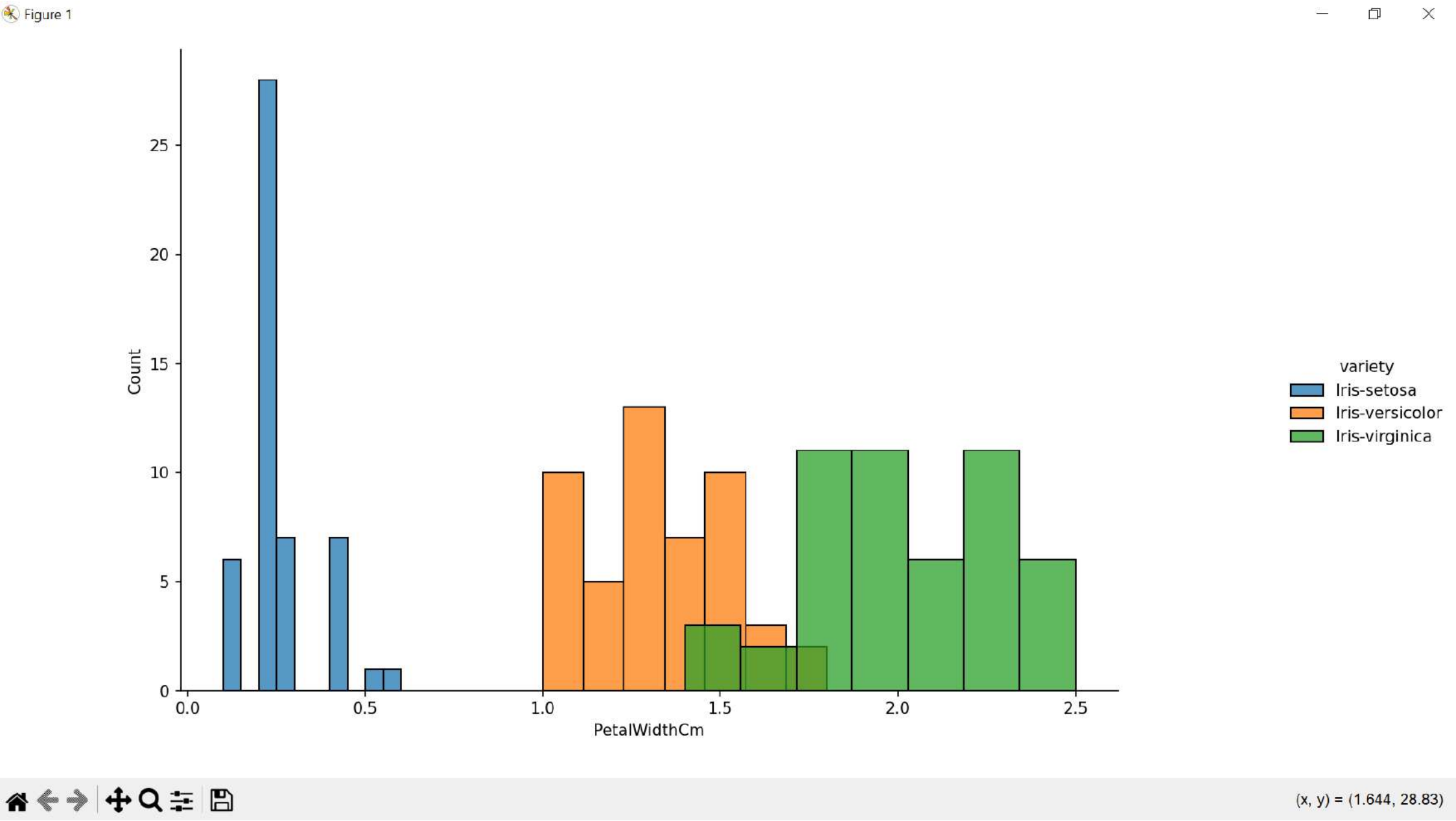

50
40
30
20
10
0
count
Iris-setosa
Iris-versicolor
Iris-virginica
variety

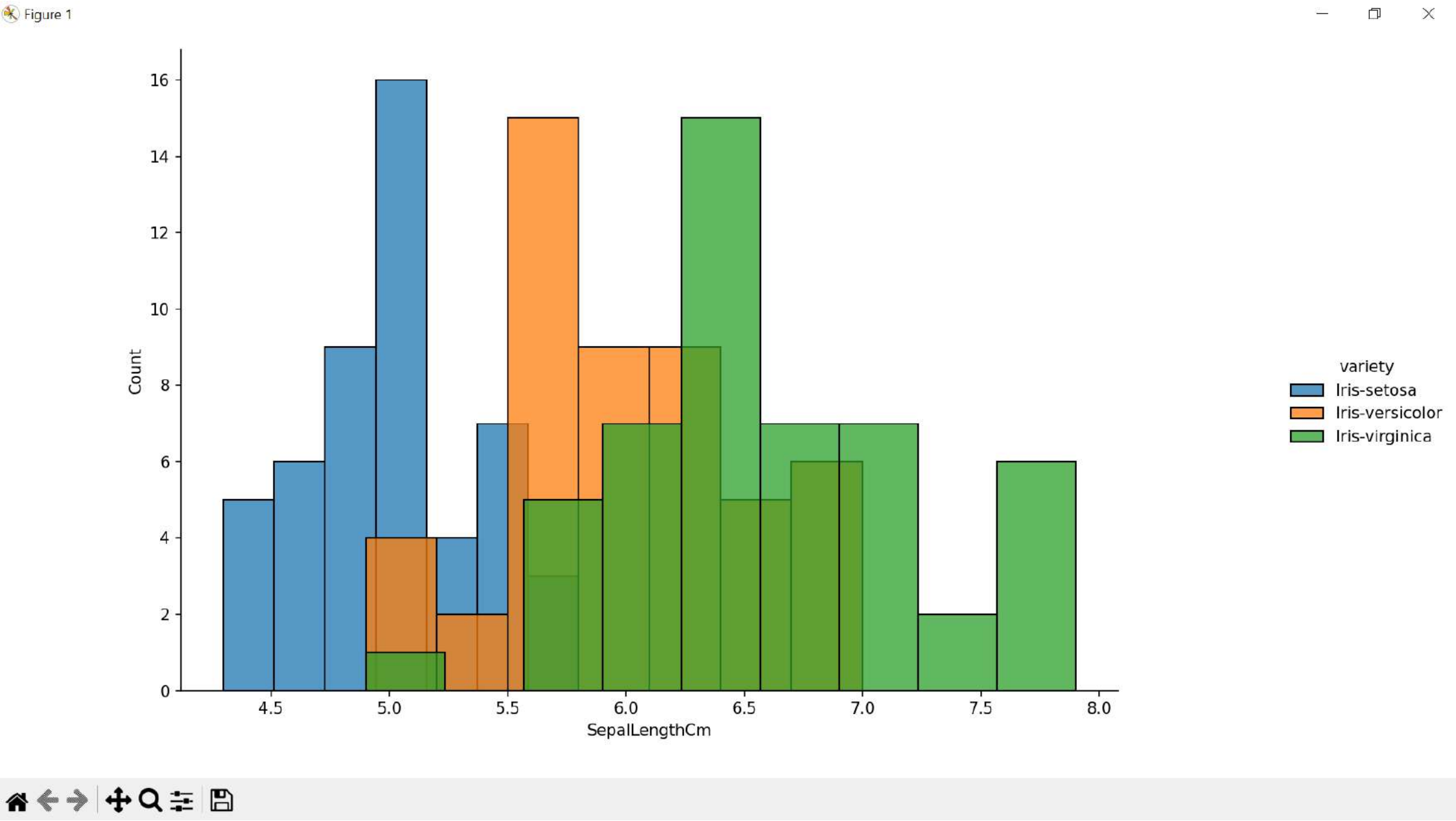

Figure 1
variety
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0

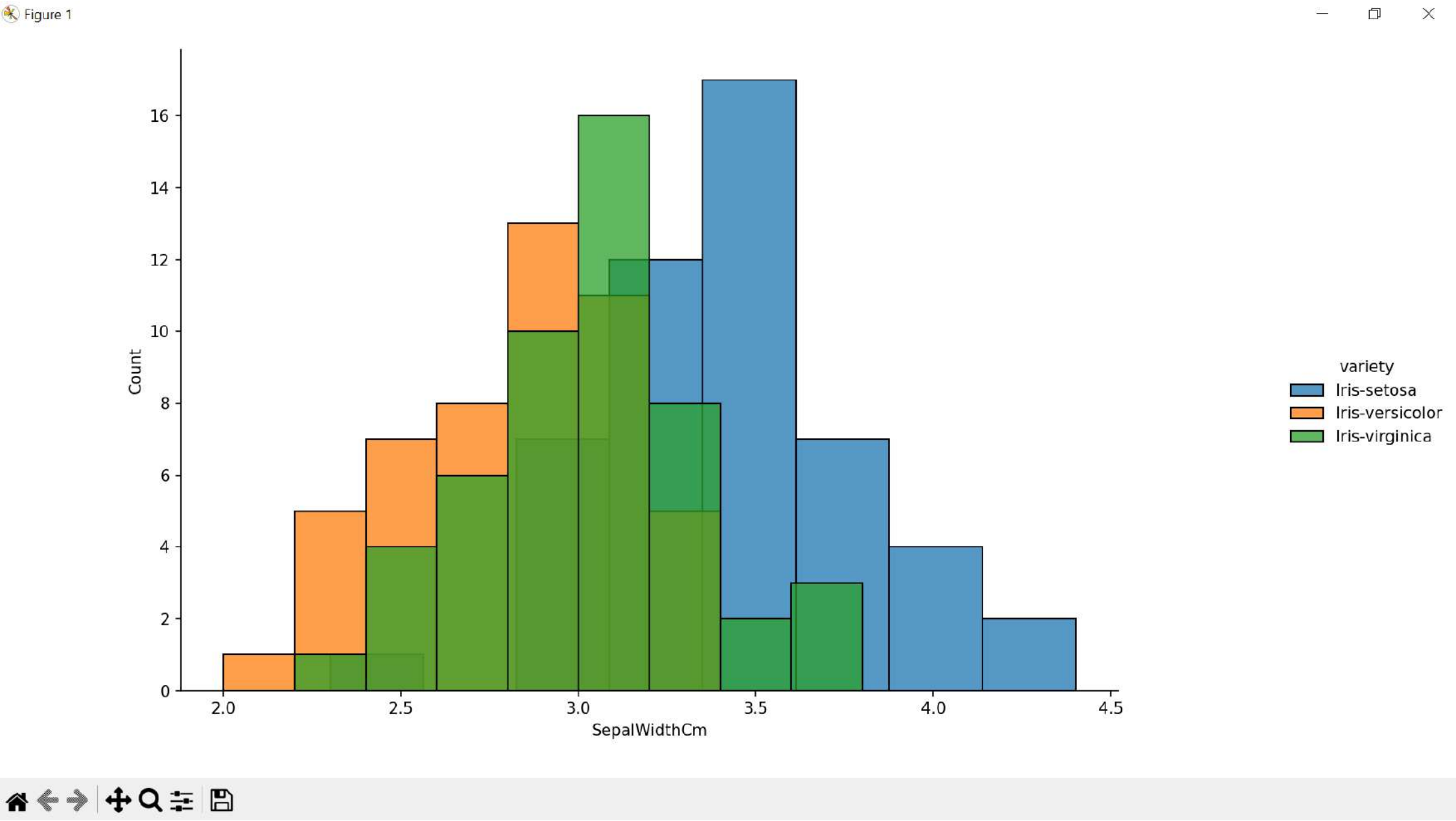

Figure 1
variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7


Figure 1
Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Figure 1
Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Figure 1
Count
PetalWidthCm
0
5
10
15
20
25
0.0
0.5
1.0
1.5
2.0
2.5
variety
Iris-setosa
Iris-versicolor
Iris-virginica
(x, y) = (1.644, 28.83)


Figure 1
16
14
12
10
8
6
4
2
0
Count
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Figure 1
Count
SepalWidthCm
0
2
4
6
8
10
12
14
16
2.0
2.5
3.0
3.5
4.0
4.5
variety
Iris-setosa
Iris-versicolor
Iris-virginica